PRACTICE SHEET-1

1. If $\frac{2p}{p^2-2p+1}=\frac{1}{4}$, then the valve of $p+\frac{1}{p}$ is
   (a) 4 (b) 5
   (c) 10 (d) 12

2. If $x^2 - 3x + 1 = 0$, then the value of $x+\frac{1}{x}$ is
   (a) 0 (b) 1
   (b) 2 (d) 3

3. If $\frac{x^2-x+1}{x^2+x+1}=\frac{2}{3}$ then the value of $x+\frac{1}{x}$ is
   (a) 4 (b) 5
   (c) 6 (d) 8

4. If $2x+\frac{1}{3x}=5$, find the value of $\frac{5x}{6x^2+20x+1}$
   (a) 1/4 (b) 1/6
   (c) 1/5 (d) 1/7

5. If $\frac{x}{x^2+3x+1}=\frac{1}{5}$, then value of $\frac{x^2+1}{x}$ is

10. If $x = 5 + 2\sqrt{6}$, then the value of $\sqrt{x}+\frac{1}{\sqrt{x}}$ is
    (a) $2\sqrt{2}$ (b) $3\sqrt{2}$
    (c) $2\sqrt{3}$ (d) $3\sqrt{3}$

11. For $a > b$, If $a + b = 5$ and $ab = 6$, then the value of $a^2 - b^2$ is
    (a) 1 (b) 3
    (c) 5 (d) 7

12. If $x-\frac{1}{x}=5$, then $x^2+\frac{1}{x^2}$ is equal to
    (a) 5 (b) 25
    (c) 27 (d) 23

13. If $x = 3 + 2\sqrt{2}$, then $\sqrt{x}-\frac{1}{\sqrt{x}}$ is equal to
    (a) 1 (b) 2
    (c) $2\sqrt{2}$ (d) $3\sqrt{3}$

14. If $x=\sqrt{3}+\sqrt{2}$, then $x^2+\frac{1}{x^2}$ is equal to